

Data Decision Log

Your Project Name Here

Your Name

January 28, 2026

Table of contents

1	Overview	3
2	Data Cleaning Decisions	3
2.1	Entry 1: Example - Handling Missing Values	3
2.1.1	Decision	3
2.1.2	Reasoning	3
2.1.3	Code Implementation	4
2.1.4	Alternatives Considered	4
2.1.5	Trade-offs	4
2.1.6	References (if applicable)	5
2.2	Entry 2: [Your Second Decision Title]	5
3	Variable Creation Decisions	5
3.1	Entry 3: [Decision About Creating/Recoding Variables]	5
3.1.1	Decision	5
3.1.2	Reasoning	5
3.1.3	Code Implementation	5
3.1.4	Alternatives Considered	6
3.1.5	Trade-offs	6
4	Analytical Decisions	6
4.1	Entry 4: [Statistical Method Choice]	6
4.1.1	Decision	6
4.1.2	Context	6
4.1.3	Reasoning	7
4.1.4	Code Implementation	7
4.1.5	Alternatives Considered	7

4.1.6	Trade-offs	7
4.1.7	Assumptions	7
4.1.8	References	7
5	Visualization Decisions	7
5.1	Entry 5: [Figure Design Choice]	7
5.1.1	Decision	8
5.1.2	Reasoning	8
5.1.3	Implementation	8
5.1.4	Alternatives Considered	8
5.1.5	Trade-offs	8
6	Status Updates & Revisions	8
6.1	Entry 6: [Revisiting a Previous Decision]	8
6.1.1	What Changed	9
6.1.2	Why the Revision	9
6.1.3	New Decision	9
6.1.4	Implications	9
6.1.5	Lessons Learned	9
7	Summary of Key Decisions	9
8	Outstanding Questions / Future Decisions	10
8.1	To Decide	10
8.2	Questions for Advisor/Collaborators	10
9	Appendix: Entry Template	10
9.1	Entry X: [Descriptive Title Goes Here]	11
9.1.1	Decision	11
9.1.2	Reasoning	11
9.1.3	Code Implementation (if applicable)	11
9.1.4	Alternatives Considered	11
9.1.5	Trade-offs	11
9.1.6	References (if applicable)	12
10	Notes on Using This Template	12
10.1	Status Badge Guide	12
10.2	When to Write an Entry	12
10.3	The Quality Test	13

1 Overview

This document records all major analytical decisions made during this project. Each entry documents what was decided, why, what alternatives were considered, and what trade-offs were involved.



How to Use This Template

1. Copy this file to your R Project folder
2. Rename it `decision_log.qmd`
3. Update the YAML header above (title, subtitle, author)
4. Delete this callout box and the example entries below
5. Start adding your own decisions using the template in the Appendix

Project start date: [Insert date]

Last updated: 2026-01-28

2 Data Cleaning Decisions

2.1 Entry 1: Example - Handling Missing Values

Date: YYYY-MM-DD

Status: Final

2.1.1 Decision

[State clearly what you decided to do]

Example: Removed all rows where the primary key variable `site_id` was NA.

2.1.2 Reasoning

[Explain WHY you made this decision. Reference documentation, theory, or practical constraints.]

Example: Cannot link observations to site characteristics without site identifiers. Checked with field team and confirmed these 3 observations were data entry errors where the site column was accidentally left blank during transcription.

2.1.3 Code Implementation

```
# Show the code that implements your decision
# Set eval: false so it doesn't actually run in this template

# Example:
# original_data <- read_csv("data/raw_data.csv")
# nrow(original_data) # 615 observations
#
# cleaned_data <- original_data %>%
#   filter(!is.na(site_id))
#
# nrow(cleaned_data) # 612 observations
```

2.1.4 Alternatives Considered

1. **Alternative 1:** [Describe alternative approach]

Reason for rejection: [Why you didn't choose this]

2. **Alternative 2:** [Describe another approach]

Reason for rejection: [Why you didn't choose this]

Example alternatives: - Create “Unknown Site” category → Rejected because observations are unusable for site-level analysis - Try to reconstruct site from GPS coordinates → Rejected because GPS data also missing

2.1.5 Trade-offs

- **Gain:** [What you gained from this decision]
- **Lose:** [What you gave up or lost]
- **Risk:** [Any risks or uncertainties]
- **Acceptable because:** [Why the trade-off is reasonable]

Example: - **Gain:** Data integrity for all site-level analyses - **Lose:** 3 observations (0.5% of total dataset) - **Risk:** Minimal - checked that excluded observations are randomly distributed across time periods - **Acceptable because:** Research questions focus on site-level patterns

2.1.6 References (if applicable)

- Citation 1
 - Citation 2
 - Discussion with [person] on [date]
-

2.2 Entry 2: [Your Second Decision Title]

Date: YYYY-MM-DD

Status: Draft

[Copy the structure from Entry 1 above]

3 Variable Creation Decisions

3.1 Entry 3: [Decision About Creating/Recoding Variables]

Date: YYYY-MM-DD

Status: Final

3.1.1 Decision

[What new variable did you create or how did you recode existing variables?]

3.1.2 Reasoning

[Why was this necessary? What does this variable represent?]

3.1.3 Code Implementation

```
# Example:
# data <- data %>%
#   mutate(
#     new_variable = case_when(
#       condition1 ~ "Category A",
#       condition2 ~ "Category B",
#       TRUE ~ "Other"
#     )
#   )
```

3.1.4 Alternatives Considered

[What other ways could you have created or defined this variable?]

3.1.5 Trade-offs

[What information is gained or lost by this categorization/transformation?]

4 Analytical Decisions

4.1 Entry 4: [Statistical Method Choice]

Date: YYYY-MM-DD

Status: Final

4.1.1 Decision

[What statistical approach or model did you choose?]

4.1.2 Context

[What problem are you trying to solve? What are the characteristics of your data?]

4.1.3 Reasoning

[Why is this approach appropriate? What assumptions are you making?]

4.1.4 Code Implementation

```
# Show the key analytical code
# Example:
# model <- lm(response ~ predictor1 + predictor2, data = data)
# summary(model)
```

4.1.5 Alternatives Considered

[What other statistical approaches did you consider?]

4.1.6 Trade-offs

[What are the strengths and limitations of this approach?]

4.1.7 Assumptions

[List key assumptions and how you checked them]

4.1.8 References

[Cite relevant statistical references or methodology papers]

5 Visualization Decisions

5.1 Entry 5: [Figure Design Choice]

Date: YYYY-MM-DD

Status: Final

5.1.1 Decision

[What type of visualization did you choose and why?]

5.1.2 Reasoning

[What does this visualization show that others don't? What makes it effective for your audience?]

5.1.3 Implementation

```
# Example visualization code
# library(ggplot2)
# ggplot(data, aes(x = variable1, y = variable2)) +
#   geom_point() +
#   labs(title = "Your title",
#         x = "X axis label",
#         y = "Y axis label") +
#   theme_minimal()
```

5.1.4 Alternatives Considered

[What other visualization types did you consider? Why didn't you use them?]

5.1.5 Trade-offs

[What does this visualization show well? What doesn't it show?]

6 Status Updates & Revisions

6.1 Entry 6: [Revisiting a Previous Decision]

Date: YYYY-MM-DD

Status: Supersedes Entry #[X]

Revision Notice

This entry supersedes [Entry X](#). The original decision has been revised based on new information or analysis results.

6.1.1 What Changed

[What decision are you revising?]

6.1.2 Why the Revision

[What did you learn that prompted this change?]

6.1.3 New Decision

[What's your updated approach?]

6.1.4 Implications

[How does this affect previous analyses? What needs to be redone?]

6.1.5 Lessons Learned

[What did you learn from this experience? This is important scientific thinking!]

7 Summary of Key Decisions

Quick Reference

Use this section to summarize the most important decisions that affect interpretation of your results:

1. [Decision 1](#): Brief description
2. [Decision 2](#): Brief description
3. [Decision 3](#): Brief description

4. [Decision 4](#): Brief description
5. [Decision 5](#): Brief description

Update this list as you add significant decisions throughout your project.

8 Outstanding Questions / Future Decisions

Use this section to track decisions you still need to make or questions you need to resolve:

8.1 To Decide

1. **[Topic needing decision]**
 - What's the question?
 - What information do you need?
 - When will you decide?
 - Who can you consult?
2. **[Another pending decision]**
 - Context
 - Next steps

8.2 Questions for Advisor/Collaborators

1. [Question 1]
 2. [Question 2]
-

9 Appendix: Entry Template

Use this template when adding new entries to your decision log. Copy and paste it, fill in the sections, and delete this appendix from your final document.

9.1 Entry X: [Descriptive Title Goes Here]

Date: YYYY-MM-DD

Status: Draft / Final / Revised

9.1.1 Decision

[One clear sentence: What did you decide?]

9.1.2 Reasoning

[Why did you make this decision? What evidence or logic supports it?]

9.1.3 Code Implementation (if applicable)

```
# Show relevant code here
# Use eval: false so template code doesn't run
```

9.1.4 Alternatives Considered

1. [Alternative 1]
Reason for rejection:
2. [Alternative 2]
Reason for rejection:

9.1.5 Trade-offs

- **Gain:**
- **Lose:**
- **Risk:**
- **Acceptable because:**

9.1.6 References (if applicable)

-
-

10 Notes on Using This Template



Tips for Success

1. **Write as you go** - Don't wait until the end to document decisions
2. **Be honest** - Uncertainty is okay and should be documented
3. **Use the test** - Could someone understand your reasoning without asking questions?
4. **Update status badges** - Draft → Final as decisions solidify
5. **Link between entries** - Use cross-references when decisions relate
6. **Render regularly** - Click "Render" to see your HTML output
7. **Commit to Git** - Track evolution of your thinking over time

10.1 Status Badge Guide

- **Draft** - Still thinking about this decision
- **Final** - Decision is made and implemented
- **Revised** - Decision changed after seeing results/new information
- **Critical** - Particularly important decision that affects key results

10.2 When to Write an Entry

Write a decision log entry when:

- You're choosing between two reasonable options
- You're combining or recoding categories
- You're excluding data (ALWAYS document exclusions)
- You're making an assumption that could be challenged
- Something didn't work and you want to remember why
- Someone might ask "why did you do it that way?"

Don't document:

- Obvious or trivial decisions
- Standard procedures followed exactly as documented
- Exploratory poking around (unless you learn something important)

10.3 The Quality Test

Could a colleague/classmate understand your reasoning without asking follow-up questions?

- If YES → You wrote enough
- If NO → Add more explanation

This decision log is version controlled with Git. Use commit messages like “Add decision about outlier handling” to track the evolution of your analytical thinking.

Remember: The thinking is the work. The code is just execution.